Zimbabwe Motor Vehicle Valuation Platform

Implementation Architecture, Valuation Methodology & User Navigation Guide

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1. Executive Overview & Problem Context

In Zimbabwe, insurers, commercial banks, asset financiers, and car buyers face a persistent challenge: there has been no universally accepted, independent, and defensible vehicle valuation standard. Values have traditionally relied on informal references or raw international book values (such as South Africa's TransUnion Auto / Dealers Guide) which fail in Zimbabwe due to import customs tariffs, multi-currency dynamics, and local parts availability.

To solve this, we have developed and integrated the Zimbabwe Motor Vehicle Valuation Platform directly into the Famba & Zimnat ecosystem. It provides a data-backed, defensible market valuation engine, a confidence scoring framework, underlying market comparables, and tamper-proof digital certificates with live QR verification.

2. Why Zimbabwe Car Pricing Differs from South Africa (SA)

The platform accounts for six critical Zimbabwean economic and automotive realities that make simple South African Rand (ZAR) conversions inaccurate:

- 1. Landed Import Dominance: Over 85% of vehicles in Zimbabwe are imported (Japan, UK, SA). Prices reflect landed CIF plus port clearance, not domestic factory MSRP.

- 2. ZIMRA Customs & Surtax: Customs duty (40% to 60%), Surtax (up to 35%), and 15% VAT mean a $3,000 import lands in Harare at $7,500 to $9,500.

- 3. The 10-Year Surtax Cliff: Vehicles older than 10 years face punitive import penalties. Consequently, clean older cars already inside Zimbabwe retain significant cash utility value.

- 4. Parts & Mechanic Ecosystem: Toyota (Hilux, Fortuner, Axio) and Honda Fit hold massive value retention due to abundant second-hand spares in Gazaland/Magaba, while European luxury cars depreciate rapidly.

- 5. Road Durability Premium: High-ground-clearance pickups (Hilux, D-Max) and SUVs command strong durability premiums due to local road conditions.

- 6. Cash USD Pricing: Vehicle transactions occur primarily in US Dollars cash, creating distinct retail vs forced liquidation spreads.

3. Architecture & What Has Been Implemented

The platform is built natively across the monorepo full stack:

A. Database Models (prisma/schema.prisma)

- VehicleValuation: Stores valuation requests, inputs (registration, VIN, mileage, condition, extras), results (low, indicative, high USD values), confidence score, sample size, methodology notes, and QR code data.

- VehicleBenchmark: Maintains base landed values, annual depreciation curves, brand durability ratings (e.g. 1.25x for Toyota), and parts liquidity rankings.

B. Valuation Engine (apps/api/src/services/valuation/)

- benchmarks.ts: Seed database covering 30+ top Zimbabwe models across Toyota, Honda, Mazda, Isuzu, Nissan, Ford, and VW.

- engine.ts: Calculates landed base replacement cost, age depreciation, mileage adjustments (+/- 1.2% per 10k km variance), condition ratings, and generates realistic local market comparables.

- Confidence Algorithm: Assigns High (sample >= 20, 90%+ match), Medium (80%+ match), or Low confidence.

- QR Code Generator: Generates cryptographic digital verification seals using qrcode.

C. Backend API Endpoints (apps/api/src/routes/valuation.ts)

- POST /api/valuation/estimate: Computes multi-factor vehicle valuation, saves to DB, returns report.

- GET /api/valuation/:idOrRef: Retrieves verified valuation record for live QR certificate checks.

- GET /api/valuation/benchmarks: Provides vehicle catalog for frontend autocomplete.

D. Web Frontend & Insurance Integration (apps/web/)

- Valuation Wizard (/valuation): Interactive 3-step valuation calculator with 1-click popular Zimbabwe model presets.

- Digital Certificate (/valuation/report/[ref]): Executive certificate with QR seal, print/PDF button, and comparable table.

- Navigation Link: Added "Vehicle Valuation" into the main Navbar.

- Insurance Sync (/insurance): Step 1 of insurance onboarding automatically connects to the valuation engine.

4. How to Navigate and Test the New Module

Follow these exact steps to view and test the valuation platform live:

Step 1: Access the Valuation Tool

- Direct URL: http://localhost:3000/valuation (or https://www.famba.co.zw/valuation)

- Main Navigation: Click on "Vehicle Valuation" in the main top navigation bar on any page.

Step 2: Try 1-Click Popular Vehicle Presets

At the top of the /valuation page, click any of the preset chips for instant pre-filling:
- Toyota Fortuner 2.8 GD-6 (2019)
- Toyota Hilux GD-6 Double Cab (2021)
- Honda Fit Hybrid GP5 (2016)
- Isuzu D-Max 3.0 Ddi (2018)
- Mazda Demio 1.3 SkyActiv (2017)

Step 3: Step-by-Step Custom Input

- Step 1 (Specs): Enter Make, Model, Variant, Year of Manufacture, Transmission, and Registration plate.

- Step 2 (Condition): Specify Odometer (e.g. 85,000 km), Condition (Excellent / Good / Fair), and Extras (Bullbar, Canopy).

- Step 3 (Purpose): Select primary purpose (Zimnat Insurance Quote, Bank Loan, Buying, or Selling) and click Generate.

Step 4: View the Digital Valuation Certificate

The system generates an official report at /valuation/report/[ref] displaying:

- Indicative Market Value (e.g. $31,500 USD) and Low/High range.
- High Confidence Score badge (94%) with verified sample size.
- Underlying Zimbabwe Market Comparables table (Borrowdale, Msasa, Bulawayo).
- Scannable QR Code seal for authenticity verification.
- "Print / Save PDF" button for official paper records.
- "Get Zimnat Quote" button that seamlessly ports the value into the insurance flow.